

In this section we discuss the notion of a limit for multivariable functions. In a single-variable context, we were content discussing *two-sided* limits since 2-dimensional graphs only had a left- and a right-hand side. When considering, for example, the surface of a function $z = f(x, y)$, the ‘directions’ become infinite.

Definition. Given a real-valued bivariate function $z = f(x, y)$ with domain D_f in $\mathbb{R}^2$, if (x_0, y_0) is taken from the interior or the boundary of D_f , then we say

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = L$$

if the values of $f(x, y)$ are made to be arbitrarily close to L as (x, y) tends to (x_0, y_0) *from any path in the interior of D_f* .

All of our favorite limit-laws from single-variable calculus still hold true in the multi-variable setting. As a result, the notion of ‘direct substitution’ will still work as long as nothing ‘goes wrong’.

Example. Compute the following limits:

$$(i) \quad \lim_{(x,y) \rightarrow (8,-8)} \frac{5x^2 - 5y^2}{x + y}$$

$$(ii) \quad \lim_{(x,y) \rightarrow (9,0)} \frac{\sqrt{9x - y} - 9}{9x - y - 81}$$

$$(iii) \quad \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^{16} + y^{16})}{x^{16} + y^{16}}$$

$$(iv) \quad \lim_{(x,y) \rightarrow (0,0)} \frac{5e^{3y} \sin(-3x)}{4x}$$

In stark contrast with single-variable calculus, however, the ‘tricks’ to solve indeterminate limits are not comprehensive. A common route is to use polar coordinates. For example,

Example. Compute the following limit: $\lim_{(x,y) \rightarrow (0,0)} \frac{4xy^2}{x^2 + y^2}$.

In single-variable calculus, a 2-sided limit existed if, in part, the two 1-sided limits agreed. This same concept is true in multivariable, but we now need to check an *infinite* number of paths. This direct sort of check is not humanly possible, so we need to dig deeper into what limits *really* are and develop *more* comprehensive techniques.

With the above said, we *can* help determine whether a proposed limit *does not* exist! To do this, we need only exhibit a **single path** that doesn’t work. To do this, there are generally good paths to try first: the coordinate axes and the line $y = x$.

Example. Consider the following limit: $\lim_{(x,y) \rightarrow (0,0)} \frac{y}{x}$. Use a path along the x -axis and the line $y = x$ to draw a conclusion.

Thus, we arrive at the following ‘**Two-Path Test for Nonexistence of a Limit**’:

If a function $f(x, y)$ has different limits along two different paths, then the limit doesn’t exist.

Example. Discuss the following limit¹: $\lim_{(x,y) \rightarrow (0,0)} \frac{2x^2y}{x^4 + y^2}$

Definition. A function $f(x, y)$ is **continuous at the point** (x_0, y_0) if

(i) $f(x_0, y_0)$ exists, (ii) $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y)$ exists, and (iii) $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = f(x_0, y_0)$.

A function is **continuous** if it is continuous for every point in its domain.

Example. Discuss the continuity of the following function²:

$$f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

¹Hint: Try the path along $y = x^2$.

²Hint: Try the path along $y = 2x$